

Nisan Chhetri

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Summary

Results-driven Ph.D. Graduate (defending December 2025; Available January 2026) specializing in Computer Vision, Machine Learning, and Generative AI/LLMs. Proven expertise in the full ML lifecycle—from feature engineering on high-dimensional data to deploying production-grade systems with multi-modal inputs (images, video, sensor data). Proficient in PyTorch, CUDA, and Cloud platforms. Available for frequent travel and on-site collaboration. Seeking a challenging full-time ML Engineer role.

Technical Skills

Languages: Python, C/C++, R, CUDA, Bash/Linux/Unix, SQL

ML/AI: PyTorch, TensorFlow, scikit-learn, Time-Series Analysis, CNNs, LSTMs, Transformers/LLMs, Diffusion Models, GANs, Computer Vision, Hugging Face, FastAPI, Streamlit, Gradio, OpenCV

Data/MLOps/Cloud: A/B Testing, Pandas, NumPy, Tableau; Docker, Kubernetes, Github; AWS, Azure, Google Cloud Platform

Education

Ph.D. in Computer Science 08/2021 – 12/2025 (expected)
North Carolina State University, Raleigh, NC

M.S. in Computer Science 08/2021 – 05/2024
North Carolina State University, Raleigh, NC

B.E. in Electrical Engineering 08/2014 – 12/2018
Tribhuvan University, Kathmandu, Nepal

Work Experience

Graduate Research Assistant, Knowledge Discovery Lab, Raleigh, NC 08/2021 – Present

- Performed R&D and prototyped a novel Generative AI framework to estimate image creativity using Novelty & Surprise.
- Executed data acquisition and data cleansing by creating a dataset via 4 MTurk studies (50+ participants, 1200+ human ratings).
- Built a scalable, automated process by integrating human feedback with LLMs to generate creativity dimension scores.
- Demonstrated leadership and teamwork by mentoring 2 undergraduate researchers; team secured 3rd place out of 15 groups.

Machine Learning Research Intern, U.S. Department of Agriculture, Beltsville, MD 05/2023 – 08/2024

- Implemented a robust, end-to-end ML pipeline to handle large-scale, complex microbiome datasets (over 4,000 features).
- Identified 3 fungal taxa predictive of pig growth with 70% accuracy, results published in the *Journal of Animal Science*.

Graduate Teaching Assistant, North Carolina State University, Raleigh, NC 08/2021 – Present

- Mentored 8 project teams, delivered 3 talks on Generative AI, and supported over 200 students across AI, C, and Discrete Math.

Machine Learning Engineer, Fusemachines Inc., Kathmandu, Nepal 12/2019 – 07/2021

- Developed and containerized an intrusion detection system (1 TB), achieving 89% accuracy; saved \$1M+ via automation.
- Prepared 10 curriculum chapters and 5 projects on RL, ML, CV, GANs, DL; mentored 3 interns across 4 colleges.

Artificial Intelligence Intern, Paaila Technology, Kathmandu, Nepal 03/2019 – 11/2019

- Improved chatbot accuracy by 6% with a Nepali-Romanizer tool, gaining 3 new enterprise clients.

Academic Projects

PromptIQ | Python, Gradio, PyTorch, GPT, Colab 05/2024 – 04/2025

- Engineered an end-to-end Computer Vision pipeline by introducing a novel CAS metric and achieved an automated visual flaw detection accuracy by nearly 240% (0.54 vs. 0.16) over the CLIP baseline across four vehicle categories.

BERT Algorithms Evaluation | Python, PyTorch, CUDA, Transformers (BERT) 08/2022 – 12/2022

- Benchmarked 5 BERT variants across 8 GLUE tasks; identified BERT Uncased as top performer on accuracy and F1.

Few-shot Energy Detection | Python, PyTorch, CUDA 01/2023 – 05/2023

- Built a few-shot model for aerial imagery of nuclear, solar, and hydro plants; achieved 95% benchmark and 58% custom accuracy.

EcoNet Weather Forecasting | Python, Scikit-learn, PyTorch 02/2022 – 05/2022

- Achieved 99% recall with CatBoost; ranked 1st of 25 teams, outperforming Random Forest, SVM, and LSTM baselines.

Publications & Awards

Publications in *Journal of Animal Science*, *Frontiers in Microbiology*, and *arXiv*; 1 under review (NeurIPS Creative AI).

USDA Research Fellowship (2023 & 2024), USDA Travel Fellowship (2023 & 2024), NC State Summer Graduate Fellowship (2022).